

GPVN -12kV Indoor AC High Voltage Vacuum Circuit Breaker

Summary

GPVN -12kV indoor AC high voltage vacuum circuit breaker is a type of holistic and completely sealed High Voltage switchgear, it has compact dimension and small volume. The VCB applies to power system of rated voltage 7.2~12kV, three-phase and AC 50/60Hz. It is applicable in occasions with frequent operations, acting to protect and control line and electric apparatus. The product conforms to IEC62271-100, GB1984, JB3855 and DL1403 and passes type test.

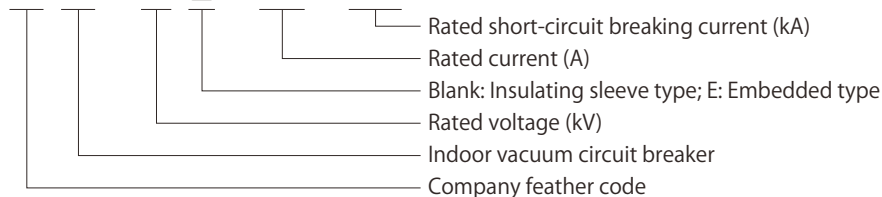
Ambient condition

Ambient temperature: -25°C ~ +45°C , daily average: ≤ 35°C
 Altitude: 1000m(Standard); can up to 4500m for special ordering;
 Relative humidity: daily average ≤ 95%, monthly average ≤ 90%;
 Earthquake intensity: ≤ 8 degree;
 Applicable occasions should be free from inflammables, explosives, corrosives and severe vibration.



Model and its meaning

GP VN - 12 □ / 630 - 31.5



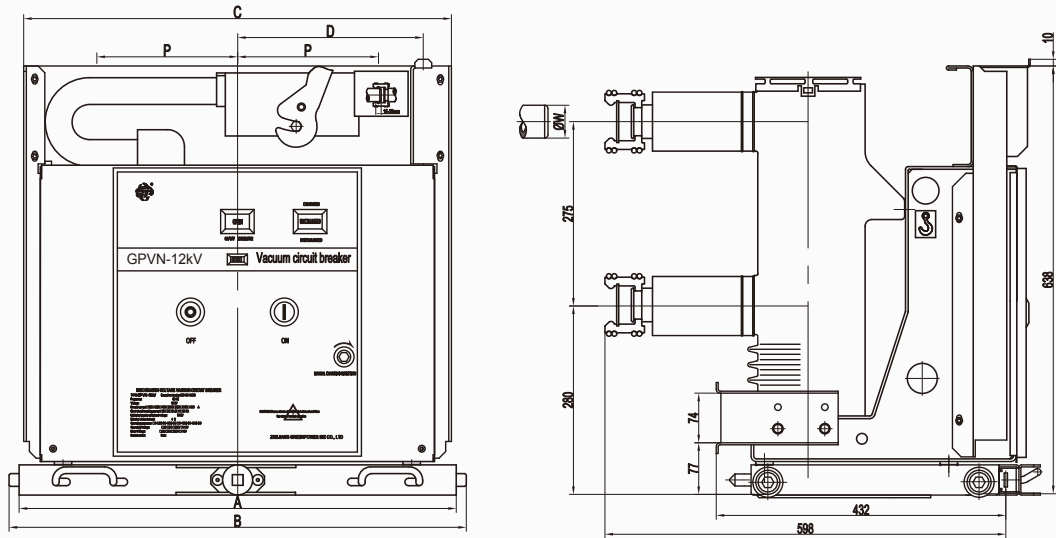
Product feature

- 1) Holistic structure
- 2) Sealed spark chamber
- 3) Effective vacuum arc-extinguish
- 4) Good mechanical behavior
- 5) Various mounting approach

Technical specifications

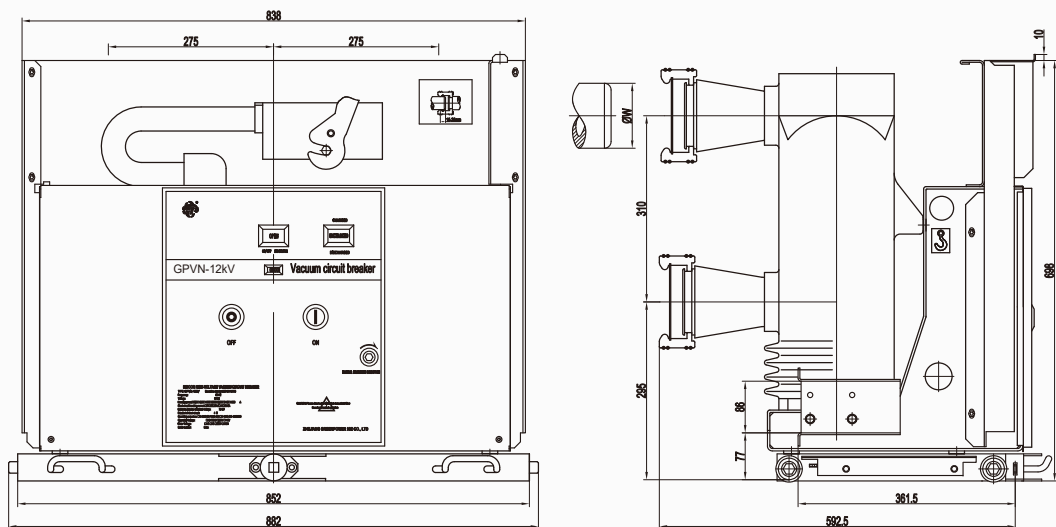
No.	Item	Unit	Data						
1	Rated voltage	kV	7.2~12						
2	Rated frequency	Hz	50/60						
3	Rated current	A	630, 1000, 1250		1600	2000, 2500, 3150		4000	
4	Rated short-circuit breaking current	kA	20	25	31.5	31.5	40	40	50
5	Rated short-circuit making current	kA	50	63	80	80	100	100	125
6	Rated short-time withstand current	kA	20	25	31.5	31.5	40	40	50
7	Rated peak withstand current	kA	50	63	80	80	100	100	125
8	Rated pull-out breaking current	kA			12.6	12.6	16	16	20
9	Rated out-of-phase earthing fault breaking current	kA	17.4	21.7	27.4	27.4	34.7	34.7	43.5
10	Rated short-circuit breaking current breaking time	times	50				30	30	12
11	Rated operation sequence		O-0.3s-CO-180s-CO				O-180s-CO-180s-CO		
12	Rated breaking current of single capacitor bank	A	630						
13	Rated breaking current of back to back capacitor bank	A	400						
14	1min Power frequency withstand voltage	kV	Phase to phase/ to earth: 42, across open contacts: 48						
15	Lightning impulse withstand voltage	kV	Phase to phase/ to earth: 75, across open contacts: 85						
16	Mechanical life	times	20000						
17	Closing time	ms	≤ 75						
18	Opening time	ms	≤ 60						
19	Power storage time under rated voltage	s	≤ 10						

■ Outline dimension of GPVN-12kV



Rated current	Rated Short Circuit Breaking Current	P	A	B	L	D	W
630A	25kA/31.5kA	150	502	532	490	202	Φ 35
		210	652	682	638	277	
1250A	25kA/31.5kA	150	502	532	490	202	Φ 49
		210	652	682	638	277	
		275	852	882	838	377	
1600A	31.5kA/40kA	210	652	682	638	277	Φ 55
		275	852	882	838	377	

Note: Fixed type and motorized type breaker are available.

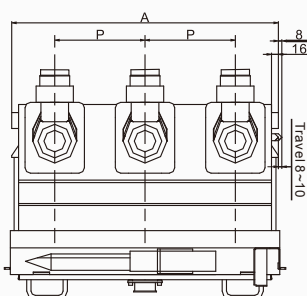
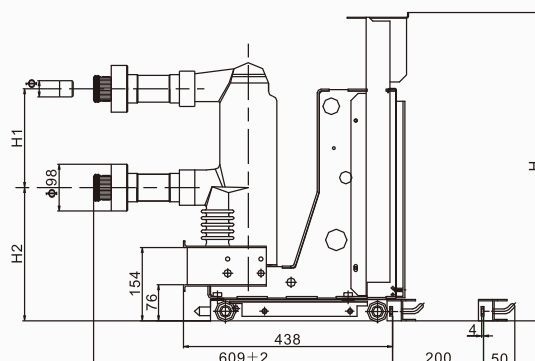
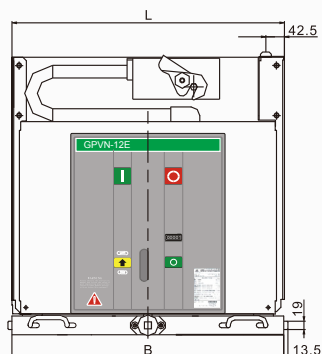


Rated current	Rated short Circuit breaking current	P	W
1600A, 2000A	31.5kA, 40kA	275	Φ 79
2500A, 3150A			Φ 109
4000A			Φ 109

Note:

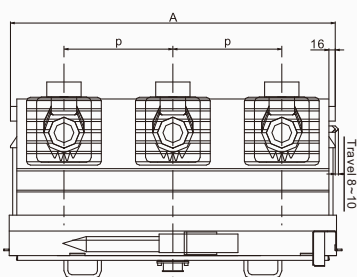
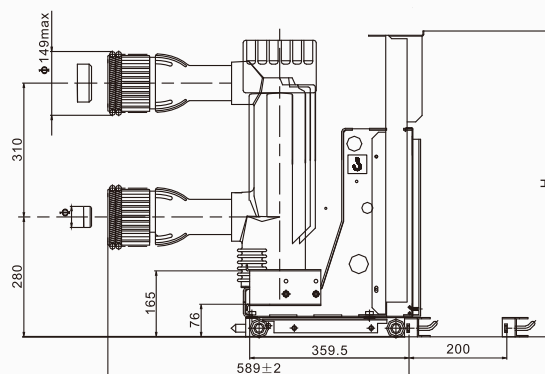
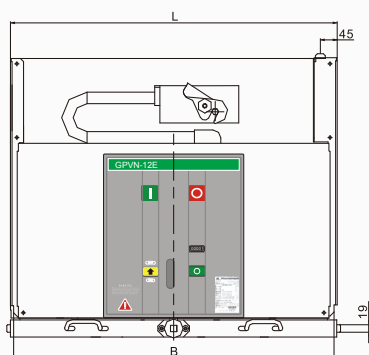
1. Fixed type and motorized type breaker are available.
2. Here only VCB schematic for reference. For the PT truck, Bus truck, fuse truck, etc. Please consult with GP for more details.

■ Outline dimension of GPVN-12E



Rated current	Rated short circuit breaking current	P	H1	H2	A	B	L	Φ	H	Applicable cabinet width	
630~1250A	20-31.5kA	150	205	260	502	502	492	35	618	650	
		210	205	260	650	652	648	35		800	
630A		210	275	280	650	652	648	35	637	800	
		150	275	280	502	502	492	35		650	
1250A		210	275	280	650	652	648	49		800	
		150	275	280	502	502	492	49		650	

Note: Fixed type and motorized type breaker are available.



Specification	P	A	B	L	Φ	H	Applicable cabinet width
1600A 31.5kA/40kA	275	850	852	844	55	694	1000
2000A 31.5kA/40kA	275	850	852	844	79	694	
2500~3150A 31.5kA/40kA	275	850	852	844	109	735	
4000A* 40kA	275	850	852	844	109	735	

Note: 1. Fixed type and motorized type breaker are available.
 2. Rated current upper than 2500A, breaker must with heat shield
 3. * Use forced air cooling, rated current can up to 4000A
 4. Here only VCB schematic for reference. For the PT truck, Bus truck, fuse truck, etc. Please consult with GP for more details.



■ Secondary diagram

